

DECATHLON

Sales & Customer Analytics

Business Requirements Document

Purpose: Define the business needs, analytical requirements, KPI definitions, dashboard requirements, user expectations, and acceptance criteria for the Excel-based Decathlon Sales & Customer Analytics solution.

Document Item	Requirement
Project	Decathlon Sales & Customer Analytics
Business Area	Retail Sales & Customer Analytics
Primary Tool	Microsoft Excel
Data Volume	30,000 synthetic retail transactions
Analysis Period	2024–2026
Primary Users	Business analysts, sales managers, category managers, decision-makers
Status	Portfolio / Demonstration Project

Based on the supplied Decathlon-style synthetic retail dataset and dashboard.

1. Business Background

The project simulates a retail analytics environment in which management needs a consolidated view of sales, customers, products, sports categories, channels, and customer behavior. The transaction data requires structured analysis and visualization before it can be used effectively for decision-making.

The proposed dashboard provides a single analytical view that reduces manual reporting effort and enables users to move from high-level KPIs to category, time, customer, and channel insights.

2. Business Problem Statement

Retail transaction data can become difficult to interpret when sales, customer, product, and operational information is spread across rows and multiple workbook sheets. Without a centralized dashboard, decision-makers may spend unnecessary time calculating KPIs and comparing performance.

The business therefore requires an interactive Excel dashboard that answers the most important commercial questions quickly and consistently.

3. Project Objectives

ID	Objective	Expected Outcome
OBJ-01	Monitor sales performance	Track sales, orders, AOV, and profit at a glance.
OBJ-02	Identify revenue drivers	Understand high-performing sports and product categories.
OBJ-03	Analyze customer behavior	Measure customer base, retention, segments, and contribution.
OBJ-04	Analyze time trends	Identify monthly and annual sales patterns.
OBJ-05	Evaluate channels	Compare website, store, and mobile-app performance.
OBJ-06	Support decisions	Convert data patterns into actionable business insights.

4. Stakeholders & User Needs

Stakeholder	Information Need	Primary Use
Sales Manager	Sales, orders, AOV, trends	Monitor commercial performance.
Category Manager	Sport and category sales	Assortment and merchandising decisions.
Marketing Manager	Customer segments, retention, promotions	Campaign and retention planning.
Store / Channel Manager	Channel and store performance	Channel optimization.
Business Analyst	Detailed transaction data and KPIs	Analysis, validation, reporting.
Management	Executive KPIs and key trends	Fast decision-making.

5. Scope

In Scope

Area	Requirement
Data	Analyze the supplied 30,000-row synthetic transaction dataset.
Sales	Calculate and display total sales, orders, AOV, and profit.
Products	Analyze product categories and sport types.
Customers	Analyze customers, segments, retention, and gender contribution.
Time	Provide annual and monthly trend analysis.
Channels	Compare website, store, and mobile app.
Dashboard	Provide interactive selectors and executive visualizations.

Out of Scope

Excluded Area	Reason
Live transactional integration	Dataset is static and synthetic.
Real customer identities	No real-world customer information is used.
Forecasting / ML	The project focuses on descriptive and diagnostic analytics.
Enterprise data warehouse	Solution is designed as an Excel portfolio project.
Production deployment	No enterprise BI deployment is required.

6. Functional Requirements

ID	Requirement	Priority
FR-01	Display total sales as a headline KPI.	High
FR-02	Display total order count.	High

FR-03	Display unique customer count.	High
FR-04	Display average order value.	High
FR-05	Display total profit and profit margin.	High
FR-06	Show top-performing sports.	High
FR-07	Show sales by product category.	High
FR-08	Show monthly sales trend.	High
FR-09	Show annual sales performance.	Medium
FR-10	Show customer retention performance.	High
FR-11	Show sales contribution by gender.	Medium
FR-12	Show sales by sales channel.	Medium
FR-13	Allow year-based filtering.	High
FR-14	Allow month-based filtering.	High

7. KPI & Metric Requirements

KPI	Business Definition	Calculation / Logic
Total Sales	Value used for executive sales reporting.	SUM(Final_Amount)
Total Orders	Number of unique orders.	DISTINCT COUNT(Order_ID)
Total Customers	Number of unique customers.	DISTINCT COUNT(Customer_ID)
Average Order Value	Average sales value per order.	Total Sales ÷ Total Orders
Total Profit	Profit generated across transactions.	SUM(Profit)
Profit Margin	Profit relative to final sales.	Total Profit ÷ Total Sales
Retention Rate	Share of customers classified as retained.	Dashboard retention logic
Return Rate	Share of transactions marked returned.	Returned ÷ Transactions
Average Rating	Average recorded customer rating.	AVERAGE(Customer_Rating)

Every KPI should use a consistent business definition throughout the workbook. KPI labels, calculation logic, and filters must remain aligned so users do not see conflicting values between the dashboard, insight sheet, and source calculations.

8. Analytical Requirements

ID	Area	Requirement
AR-01	Time Analysis	Compare sales by year, month, quarter, and monthly trend.
AR-02	Sports Analysis	Rank sports by sales and order activity.
AR-03	Category Analysis	Measure sales contribution by product category.
AR-04	Customer Analysis	Review customer count, segments, retention, and gender.
AR-05	Channel Analysis	Compare website, store, and mobile app performance.
AR-06	Operational Analysis	Use delivery, returns, and ratings where relevant.
AR-07	Decision Support	Translate significant patterns into business recommendations.

9. Dashboard Requirements

Component	Requirement
Executive KPI Area	Prominently display Total Sales, Orders, Customers, AOV, Profit, and related metrics.
Top Sports	Display the five highest-performing sports with sales and order information.
Sales Trend	Show monthly sales movement across the selected period.
Category Analysis	Use a clear visual to communicate category contribution.
Retention	Display overall and/or monthly customer retention.

Gender Analysis	Compare sales contribution by gender.
Channel Analysis	Show sales contribution by channel.
Filters	Provide Year and Month selectors.
Navigation	Keep the dashboard easy to scan with clear hierarchy and labels.

10. Data Requirements

Data Domain	Required Fields
Order	Order_ID, Order_Date, Quantity
Customer	Customer_ID, Customer_Name, Gender, Age, Age_Group, Customer_Segment, Membership
Product	Product_ID, Product_Name, Product_Category, Brand, Sport_Type
Sales	Unit_Price, Discount_Percent, Discount_Amount, Sales_Amount, Final_Amount, Cost_Price, Profit
Store	Store_ID, Store_Name, City, State
Channel	Sales_Channel, Payment_Method, Salesperson
Delivery	Delivery_Type, Delivery_Days
Experience	Customer_Rating, Return_Status, Return_Reason
Marketing	Promotion_Campaign
Time	Quarter, Month, Year

11. Data Quality Requirements

ID	Requirement	Acceptance Condition
DQ-01	Unique order identifiers	Order IDs support accurate order counting.
DQ-02	Customer identifiers	Customer IDs support unique-customer analysis.
DQ-03	Date validity	Order dates are valid for time analysis.
DQ-04	Numeric sales fields	Sales, price, discount, and profit fields are numeric.
DQ-05	Categorical consistency	Category, sport, gender, and channel labels are standardized.
DQ-06	KPI reconciliation	Dashboard totals reconcile with source calculations.

12. Non-Functional Requirements

Requirement	Expected Standard
Usability	Business users should understand the dashboard with minimal training.
Performance	Filters and dashboard interactions should remain responsive in Excel.
Accuracy	Displayed KPIs must reconcile with approved source calculations.
Consistency	Fonts, labels, number formats, and visual hierarchy remain consistent.
Readability	Charts use clear titles, units, legends, and appropriate scales.
Maintainability	Source data and calculations are organized for future updates.
Portability	Workbook opens correctly in a current version of Microsoft Excel.

13. Business Rules

Rule	Business Logic
BR-01	Final_Amount is the primary sales value used for executive reporting.
BR-02	Orders are counted using unique Order_ID values.
BR-03	Customers are counted using unique Customer_ID values.
BR-04	AOV equals total sales divided by unique orders.
BR-05	Profit margin equals profit divided by final sales.
BR-06	Return rate is based on transactions marked Returned.
BR-07	Year and Month filters update relevant dashboard visuals.
BR-08	Synthetic data must not be presented as official Decathlon business data.

14. User Stories

ID	User Story
US-01	As a sales manager, I want to see total sales and orders so that I can quickly assess performance.
US-02	As a category manager, I want to rank sports and categories so that I can identify revenue drivers.
US-03	As a marketing manager, I want retention and customer segments so that I can plan lifecycle campaigns.
US-04	As a channel manager, I want to compare sales channels so that I can prioritize channel investments.
US-05	As a business analyst, I want Year and Month filters so that I can investigate trends.
US-06	As management, I want an executive summary so that I can make decisions without reviewing raw transactions.

15. Acceptance Criteria

ID	Acceptance Criterion
AC-01	Dashboard displays the agreed headline KPIs.
AC-02	Total sales reconciles to the source Final_Amount field.
AC-03	Total orders reconciles to unique Order_ID values.
AC-04	Total customers reconciles to unique Customer_ID values.
AC-05	AOV equals Total Sales divided by Total Orders.
AC-06	Top sports and categories are correctly ranked by the selected sales metric.
AC-07	Year and Month selectors update relevant dashboard visuals.
AC-08	Charts have meaningful titles, labels, and readable number formats.
AC-09	Dashboard is usable without modifying formulas or source data.
AC-10	Workbook clearly identifies the dataset as synthetic.

16. Assumptions & Constraints

Type	Statement
Assumption	The supplied workbook is the approved source for this portfolio project.
Assumption	Final_Amount is the preferred sales value for dashboard reporting.
Assumption	The dashboard is intended for descriptive and diagnostic analysis.
Constraint	Data is synthetic and may not reflect real Decathlon purchasing behavior.
Constraint	The project is implemented in Excel rather than a production BI platform.
Constraint	No live data refresh or enterprise authentication is required.

17. Risks & Mitigations

Risk	Impact	Mitigation
Synthetic data bias	Insights may not represent real customer behavior.	Clearly label data as synthetic.
KPI definition drift	Different sheets could show conflicting values.	Document and standardize KPI formulas.
Manual Excel changes	Formulas or visuals may be altered.	Separate source, calculations, and dashboard areas.
Workbook growth	Future data may reduce Excel performance.	Use structured tables or consider Power BI at scale.
Ambiguous retention	Retention comparisons may be misleading.	Document and consistently apply the retention rule.

18. Implementation Roadmap

Phase	Activity	Output
1. Understand	Review business questions and source workbook.	Agreed analytical scope
2. Prepare	Validate and organize transaction data.	Analysis-ready dataset
3. Calculate	Build KPIs and supporting calculations.	Validated metric layer
4. Visualize	Create dashboard charts, cards, and filters.	Interactive Excel dashboard
5. Validate	Reconcile dashboard values against source calculations.	QA-approved dashboard
6. Document	Create report, requirements, and walkthrough.	Portfolio documentation

19. Success Measures

Measure	Target
KPI accuracy	100% reconciliation with approved calculation logic.
Dashboard usability	Core business questions answerable without inspecting raw rows.
Filter functionality	Year and Month controls correctly update relevant views.
Business coverage	Sales, customer, product/sport, time, and channel requirements represented.
Documentation	Business requirements, project report, and dashboard walkthrough available.

20. Final Requirement Statement

The completed solution must provide an accurate, interactive, and easy-to-understand Excel dashboard that converts the supplied 30,000-row synthetic retail dataset into actionable sales and customer insights. The dashboard must satisfy the defined KPI, analytical, filtering, usability, data-quality, and acceptance requirements while clearly identifying the data as synthetic.

Prepared as a Business Analytics Portfolio Project

Data Disclaimer: This document uses synthetic data for educational and portfolio purposes. It does not represent official Decathlon customer, sales, financial, or operational information.